Assignment: Building a Mixture of Experts (MoE) Router

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SECTION: A

Objective

The objective of this assignment was to build a Smart Customer Support Router using a Mixture of Experts (MoE) architecture with the Groq API.

Instead of using a single general AI model to handle all types of queries, the system intelligently routes user requests to specialized experts such as:

- **Technical Expert** (for bug reports and coding issues)
- **Billing Expert** (for payment and refund-related queries)
- **General Expert** (for casual or fallback conversations)

This design simulates how real-world companies divide support responsibilities across departments.

Mixture of Experts (MoE) Concept

Mixture of Experts is an architecture where:

- A **Router model** analyzes the input and decides which expert should handle it.
- A **Specialized expert** generates the final response based on domain-specific behavior.

Instead of one model handling every task, the system becomes modular, structured, and scalable.

Implementation Details

1) Setup

- Installed `groq` and `python-dotenv`
- Loaded the `GROQ_API_KEY` securely using environment variables
- Used Groq's LLM models for both routing and response generation

This ensures secure API management and proper system configuration

2) Defining the Experts

A configuration dictionary was created to define each expert.
Each expert configuration includes: **Model name, System prompt, Temperature value**

Different system prompts were used to simulate specialization while using the same base model.

The Technical Expert is code-focused and precise, providing structured debugging guidance for programming-related issues.

The Billing Expert is empathetic and professional, focusing on financial clarity and policy-based responses.

The General Expert acts as a fallback, handling general or casual conversations when a query does not fit other categories.

Temperature settings:

- **0.7 for experts**: Allows flexible and natural responses
- **0 for routing**: Ensures consistent classification

3) Routing Mechanism

A routing component was implemented to classify user intent into predefined categories.

The router:

- Sends the user query to the LLM
- Uses a strict classification instruction
- Returns only the category label
- Operates with temperature set to 0 for deterministic behavior

This ensures stable and reliable decision-making.

4) Orchestration Logic

An orchestration layer was implemented to manage the overall workflow.

The process:

1. The router determines the query category
2. The corresponding expert configuration is selected
3. The expert generates the response using its specialized system prompt

4. The final answer is returned to the user

This design separates decision-making from response generation, improving clarity and modularity.

Bonus: Tool Integration

A tool-based category was added to extend functionality.

For specific types of requests, the system routes the query to a predefined function instead of generating a standard LLM response.

This demonstrates how LLM-based systems can integrate deterministic tools or external data sources, making the architecture more practical and extensible.

Conclusion

This project demonstrates an understanding of the Mixture of Experts architecture and its practical implementation using the Groq API.

By separating routing logic from expert responses and simulating specialization through system prompts, the system becomes modular, scalable, and closer to real-world AI customer support architectures.